PYTHON + AI AGENTS CHEAT SHEET

◆ 1. Dictionaries (Core)

✅ Create

```
d = {"a": 1, "b": 2}
```

✅ With list inside

```
d = {"marks": [80, 90, 85]}
```

✅ Input

```
d = {}
n = int(input())

for i in range(n):
    key = input()
    value = input()
    d[key] = value
```

✅ Output

```
print(d)              # full dict
print(d["a"])         # specific value

for k in d:
    print(k)          # keys

for v in d.values():
    print(v)          # values

for k, v in d.items():
    print(k, v)       # key + value
```

⚠□ Key Rule

- Keys → immutable (str, int)
- Values → anything (list, dict, etc.)

◆ 2. `.items()` Concept

```
for k, v in d.items():
```

👉 Returns:

```
(key, value)
```

Example:

```
("room1", "dirty")
```

◆ 3. Accessing Values (Percept)

```
d[key]
```

👉 Returns value of that key

✅ Safe version:

```
d.get(key, "Not found")
```

◆ 4. Dictionary Update (VERY IMPORTANT)

```
d[key] = new_value
```

⚠️ Key must match exactly:

```
"Room1" ❌
"Room 1" ✅
```

◆ 5. Simple Reflex Agent

✅ Idea:

- No memory ❌
- Only current state ✅

```
if percept == "occupied":
    action = "ON"
else:
    action = "OFF"
```

✅ Example:

```
if rooms[room] == "occupied":
    print("LIGHT ON")
```

◆ 6. Model-Based Agent

✅ Idea:

- Has memory ✓□
- Tracks past ✓□

```
self.checked = []
self.collected = []
```

◆ 7. Environment + Agent Structure

```
class Environment:
    def get_percept(self):
        return state

class Agent:
    def act(self, percept):
        return action
```

◆ 8. Utility-Based Agent ⭑ (VERY IMPORTANT)

✅ Utilities Dictionary

```
utilities = {
    "move": 50,
    "stop": 20
}
```

✅ Best Action

```
best_action = max(utilities, key=utilities.get)
```

☞ Picks action with **highest value**

❌ Wrong

```
max(utilities)   # compares keys
```

◆ 9. Random Values (Environment Simulation)

```
import random

random.randint(1,10)
random.choice(["a","b","c"])
```

◆ 10. Common Errors (YOU MADE THESE 👀)

❌ Undefined variable

```
for x in shelves   # wrong
```

✅

```
for x in self.shelves
```

❌ Wrong attribute

```
self.checkshelves
```

✅

```
self.checkedshelves
```

❌ Wrong object

```
shelf.checkedshelves.append()
```

✅

```
self.checkedshelves.append()
```

❌ Split error

```
room_name, state = input().split()
```

✅ Better:

```
room, num, state = input().split()
room_name = f"{room} {num}"
```

❌ String syntax error

`["clear,"windy"]` ❌

✓□

`["clear", "windy"]`

◆ 11. Infinite Loop (Continuous System)

```
while True:
    ...
    if condition:
        break
```

◆ 12. Key Exam Concepts Summary

Concept	Key Idea
Dictionary	key → value
`.items()`	(key, value) pairs
Reflex Agent	no memory
Model-Based	has memory
Utility Agent	chooses best action
`max(..., key=...)`	compares values
Update	key must match exactly

🔥 FINAL MEMORY TRICK

☞ AI Agent Flow

`Percept → Condition → Action`

☞ Utility Agent

`Actions → Assign scores → Pick max`